

SANJIT MATHUR

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Education & Summary

University of Wollongong in Dubai – Expected 2027

CAS Engineering student at UOW Dubai building ML systems and distributed fintech tooling. Experience deploying production services on Kubernetes and developing anomaly detection, forecasting, and LLM-powered applications across fintech and aviation datasets.

Skills

Languages: Python, TypeScript, C, SQL, MATLAB

AI/ML: LLM APIs (prompt engineering, RAG), OpenCV, scikit-learn, Pandas, NumPy, Matplotlib, Stacking Ensembles, Quantile Regression

Backend: Node.js, Express.js, REST APIs, Pydantic (Data Validation)

Infrastructure & Tools: Docker, Kubernetes, PostgreSQL, Streamlit, Plotly, Joblib

Frontend: React, Next.js

Experience

AI Engineering Intern | Baraka Financial Ltd., Dubai (Feb 2026 – Present)

- Deployed containerized services to Kubernetes; built AI-powered internal tooling that automated error classification and log analysis, eliminating manual triage across support workflows.
- Built Position Search, Trading Account Monitor, CRA, EOD History, Slack Automation and other modules integrating OMS, Instruments, and Wallet microservices, reducing manual portfolio lookup time for operations teams.
- Unified inconsistent data schemas across distributed services into a single portfolio state model, eliminating data discrepancies in cross-market account views.

Digital Intern | IndiGo InterGlobe Aviation Ltd., Gurgaon (August 2025 – Sept 2025) – [GitHub](#)

- Built a Logistic Regression model to predict on-time arrival performance on the DEL–BOM sector using **1,000 flight records and 6 engineered features** (block-hour overrun, departure delay, aircraft type, weather index, ATC congestion), achieving **88% accuracy** with balanced precision/recall across both classes.
- Engineered features from raw operational data including one-hot encoding of **3 aircraft types** (A320, A320neo, A321neo), computed block-hour overrun from scheduled vs. actual hours, and conducted correlation analysis across **10 numerical variables** to inform feature selection for modeling.

Software Engineering Intern | Lab of Future, Dubai (Jun 2025- Aug 2025)

- Built internal educational software used by **500+ students across 4 campuses**.
 - Designed backend APIs for course content delivery and student progress tracking.
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Projects

Distributed Fraud Detection & Anomaly Analysis | [GitHub](#), [Live Website](#)

- Built an **end-to-end fraud detection system** analyzing credit card transactions using unsupervised anomaly detection models (Isolation Forest, LOF, One-Class SVM, Autoencoder).
- Engineered domain features and trained models using **normal-only data pipelines with automated threshold optimization** via precision-recall analysis.
- Developed an **interactive Streamlit dashboard with SHAP explainability** for transaction-level fraud reasoning and real-time anomaly investigation.

Multi Domain Demand Forecaster | [GitHub](#), [Live Website](#)

- Built a **machine learning forecasting engine** using **XGBoost + LightGBM** stacking ensemble with a Ridge meta-learner for multi-domain demand prediction.
- Engineered advanced temporal features including **cyclical encodings and rolling statistics** to capture seasonality and trend dynamics.
- Deployed a **FastAPI prediction service and Streamlit analytics dashboard** for real-time forecasts and model performance monitoring.

ExamForge – AI Exam Generation Platform | [GitHub](#)

- Built a full-stack AI exam generation platform using LLM APIs to dynamically generate multi-format assessments across subjects.
- Designed interactive workflows enabling teachers to generate, attempt, and review customized exams with automated evaluation pipelines.

Orvyn ExoArm – Rehabilitation Exoskeleton (Current – May 2026)

- Developing signal classification pipeline to differentiate between intended finger movements and noise, targeting **90%+ accuracy** for assisted rehab exercises.
- Designing a microcontroller-based rehabilitation exoskeleton that interprets sEMG signals to assist finger movement through real-time signal processing and controlled servo actuation.

F1 Simulation Dashboard | [GitHub](#)

- Built a full-stack race simulation engine modeling tire degradation, pit strategy, safety cars, weather, and overtaking across all **24 2026 GPs using Monte Carlo probability methods.**
- Implemented dual-mode interface: season-wide predictions and fully configurable custom race simulations.